

Assignment 4(b)

Title: Write a program for matrix multiplication using CUDA

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Div:B

Batch:B1

```
#include <cuda_runtime.h>
#include <iostream>
__global__ void matmul(int* A, int* B, int* C, int N) {
    int Row = blockIdx.y*blockDim.y+threadIdx.y;
    int Col = blockIdx.x*blockDim.x+threadIdx.x;
    if (Row < N && Col < N) {
        int Pvalue = 0;
        for (int k = 0; k < N; k++) {
            Pvalue += A[Row*N+k] * B[k*N+Col];
        }
        C[Row*N+Col] = Pvalue;
    }
}

int main() {
    int N = 512;
    int size = N * N * sizeof(int);
    int* A, * B, * C;
    int* dev_A, * dev_B, * dev_C;
    cudaMallocHost(&A, size);
    cudaMallocHost(&B, size);
    cudaMallocHost(&C, size);
    cudaMalloc(&dev_A, size);
    cudaMalloc(&dev_B, size);
    cudaMalloc(&dev_C, size);
    // Initialize matrices A and B
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++) {
            A[i*N+j] = i*N+j;
            B[i*N+j] = j*N+i;
        }
    }
    cudaMemcpy(dev_A, A, size,
               cudaMemcpyHostToDevice);
    cudaMemcpy(dev_B, B, size,
               cudaMemcpyHostToDevice);
    dim3 dimBlock(16, 16);
    dim3 dimGrid(N/dimBlock.x, N/dimBlock.y);
```

```

matmul<<<dimGrid, dimBlock>>>(dev_A, dev_B,
dev_C, N);
cudaMemcpy(C, dev_C
// Print the result
for (int i = 0; i < 10; i++) {
for (int j = 0; j < 10; j++) {
std::cout << C[i*N+j] << " ";
}
std::cout << std::endl;
}
// Free memory
cudaFree(dev_A);
cudaFree(dev_B);
cudaFree(dev_C);
cudaFreeHost(A);
cudaFreeHost(B);
cudaFreeHost(C);
return 0;

```

Output:

```

datanalytics2@datanalytics2-OptiPlex-3050:~$ nvcc matrix_mul.cu -o matrix_mul
datanalytics2@datanalytics2-OptiPlex-3050:~$ ./matrix_mul
0 512 1024 1536 2048 2560 3072 3584 4096 4608
512 1024 1536 2048 2560 3072 3584 4096 4608 5120
1024 1536 2048 2560 3072 3584 4096 4608 5120 5632
1536 2048 2560 3072 3584 4096 4608 5120 5632 6144
2048 2560 3072 3584 4096 4608 5120 5632 6144 6656
2560 3072 3584 4096 4608 5120 5632 6144 6656 7168
3072 3584 4096 4608 5120 5632 6144 6656 7168 7680
3584 4096 4608 5120 5632 6144 6656 7168 7680 8192
4096 4608 5120 5632 6144 6656 7168 7680 8192 8704
4608 5120 5632 6144 6656 7168 7680 8192 8704 9216
datanalytics2@datanalytics2-OptiPlex-3050:~$

```